

CSCI 361 Lecture 6:

Pumping Lemma

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Announcements & Logistics

- **HW 3** is due tomorrow **at 10 pm** on Gradescope
- Hand in Exercise #5, pick up Exercise #6
- Reminder: In class midterm I on Oct 7

Last Time

- Discussed Myhill Nerode theorem
 - Max fooling set size of $L = \#$ equivalence classes of $\equiv_L$
 $=$ min states of DFA for L
- Practice using it to prove some languages are not regular
 - $\{a^n b^n \mid n \in \mathbb{N}\}$
 - $\{ww \mid w \in \{0,1\}^*\}$
 - $\{a^n \mid n \in \mathbb{N} \text{ and } n \text{ is a power of } 2\}$
 - $\{w \in \{0,1\}^* \mid w \text{ has an equal number of 0s and 1s}\}$

Review of Examples

- $L = \{a^i b^i \mid i \in \mathbb{N}\}$ is not regular
 - $F = \{a^i \mid i \in \mathbb{N}\}$ is an infinite fooling set since for any pair $a^i, a^j \in F$ such that $i \neq j$, suffix $z = b^i$ is such that $az \in L$ and $bz \notin L$
- $L = \{a^n \mid n \in \mathbb{N} \text{ and } n \text{ is a power of } 2\}$ is not regular
 - $F = \{a^{2^i} \mid i \in \mathbb{N}\}$ is an infinite fooling set
- $L = \{ww \mid w \in \{0,1\}^*\}$ is not regular
 - $F = \{0^i 1 \mid i \in \mathbb{N}\}$ is an infinite fooling set

Review of Closure Properties

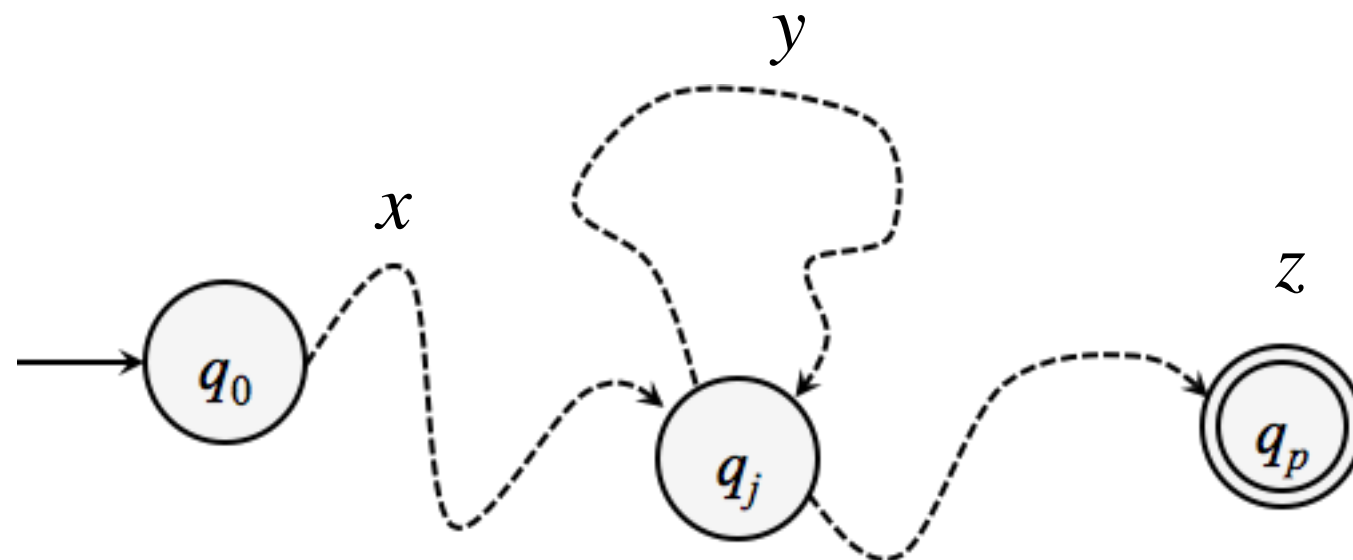
- Can often use closure properties to argue non-regularity
- $L = \{w \in \{0,1\}^* \mid w \text{ has an equal number of 0s and 1s}\}$ is not regular
- We know $L' = \{0^n 1^n \mid n \in \mathbb{N}_0\}$ is not regular (proved in class)
- Suppose L is regular, then since regular languages are closed under concatenation, we know $L \cap 0^* 1^*$ must be regular
- But $L' = L \cap 0^* 1^*$ which is a contradiction

Today: Pumping Lemma

- A necessary (but not sufficient) condition for regular languages
- Weaker guarantee but will generalize to the next class of languages

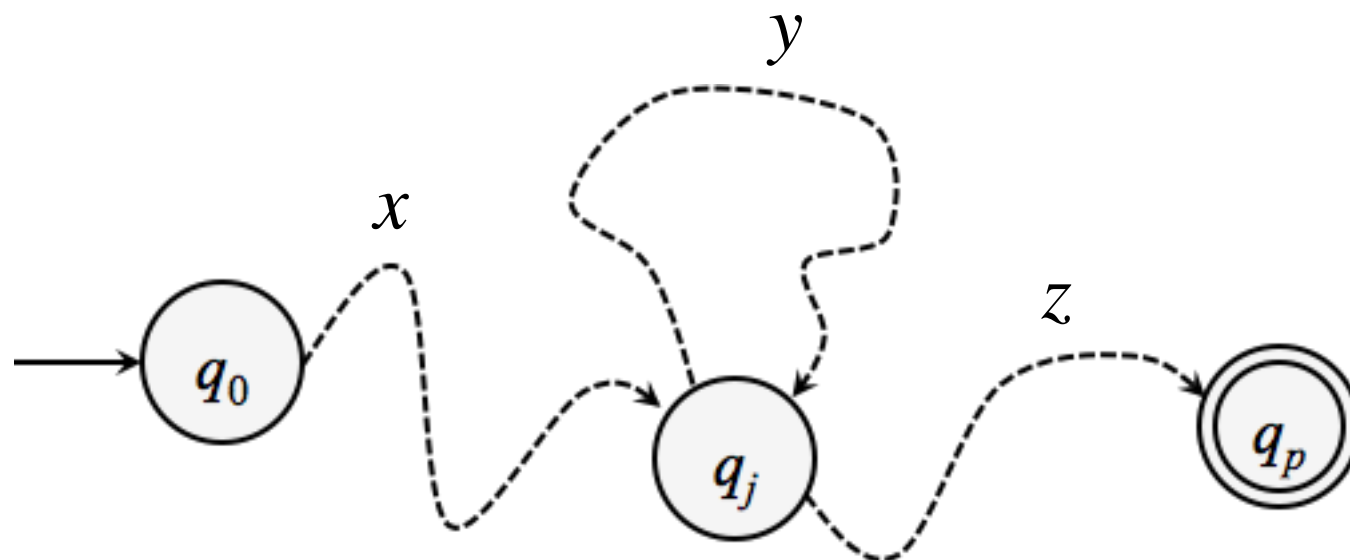
Pumping Lemma: Intuition

- If DFA M has p states then M visits a state more than once on any string with length at least p
 - Number of states visited = length of string + 1
- Let $w = xyz$ be the string that is accepted such that y is component in between the first repeated state (q_j)
 - Then xy^iz should also be accepted (can "pump" the middle piece repeatedly)



Pumping Lemma: Proof

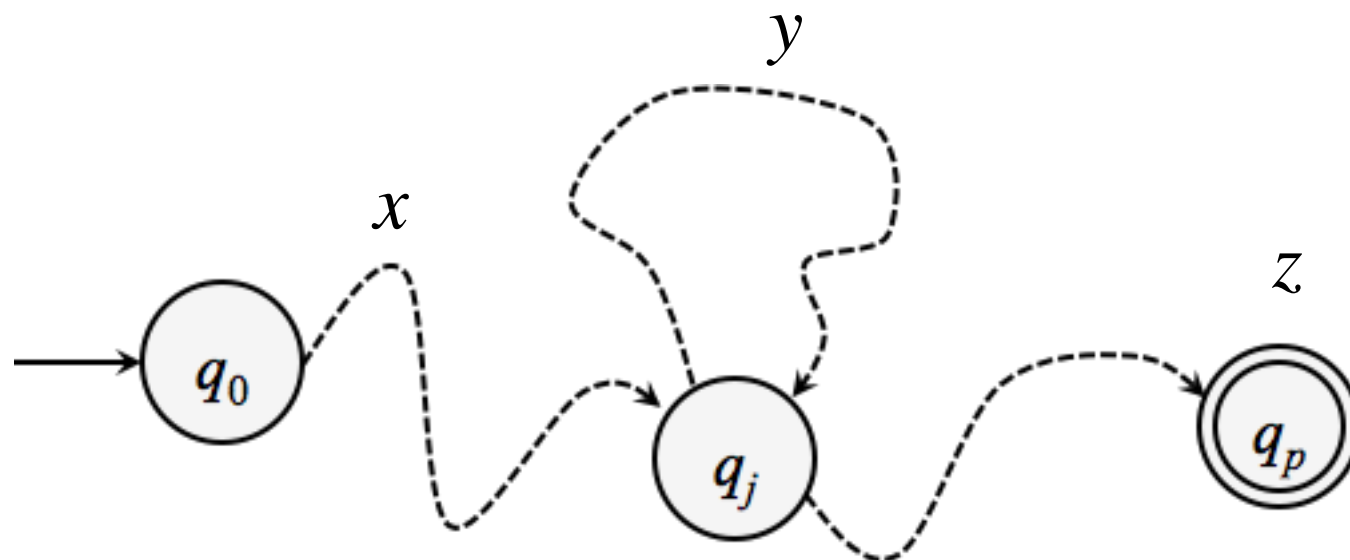
- Consider DFA M for L . Let p be the number of states in M
- Let s be a string of length $n \geq p$
- Then M 's computation sequence enters $n + 1$ states on s
- By pigeonhole principle, there must be a repeated state q_j in the first $p + 1$ states of this sequence
- Let x be the substring that brings M from q_0 to first occurrence of q_j



Formal Statement

Pumping Lemma. If L is a regular language, then there exists a number p where if $w \in L$ is any string of length at least p , then w may be divided into three pieces $w = xyz$ such that:

1. $|y| > 0$
2. $|xy| \leq p$ (y must appear amongst the first p symbols)
3. for each $i \geq 0$, $xy^iz \in L$



Pumping Lemma: Game View

- Defender claims L satisfies pumping lemma
- Challenger claims L does not satisfy pumping lemma

Defender

Pick pumping length p

Divide s into xyz

s.t. $|y| > 0$ and $|xy| \leq p$

Challenger

Pick $s \in L$ s.t. $|s| \geq p$

Pick i , such that $xy^iz \notin L$

$\xrightarrow{p}$
 $\xleftarrow{z}$
 x, y, z
 $\xrightarrow{\quad}$
 $\xleftarrow{i}$

Pumping Lemma: Game View

- If L is regular: defender has a winning strategy, challenger gets stuck
- If challenger has a winning strategy, L cannot be regular

Defender

Pick pumping length p

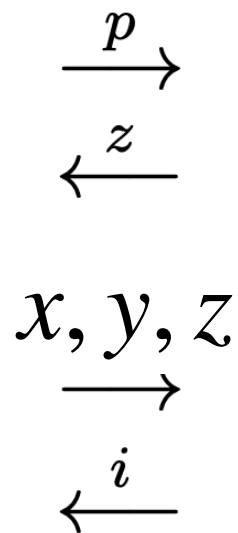
Divide s into xyz

s.t. $|y| > 0$ and $|xy| \leq p$

Challenger

Pick $s \in L$ s.t. $|s| \geq p$

Pick i , such that $xy^iz \notin L$



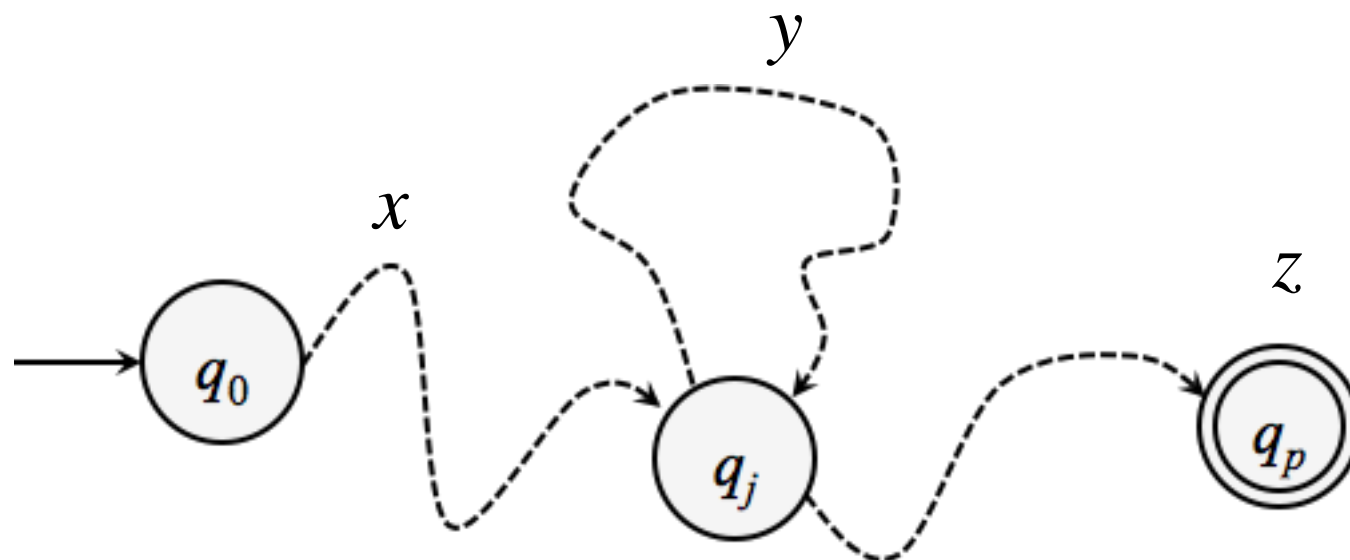
Questions

- Do all regular languages satisfy the pumping lemma?
- If a language satisfies the pumping lemma, does that mean it is regular?

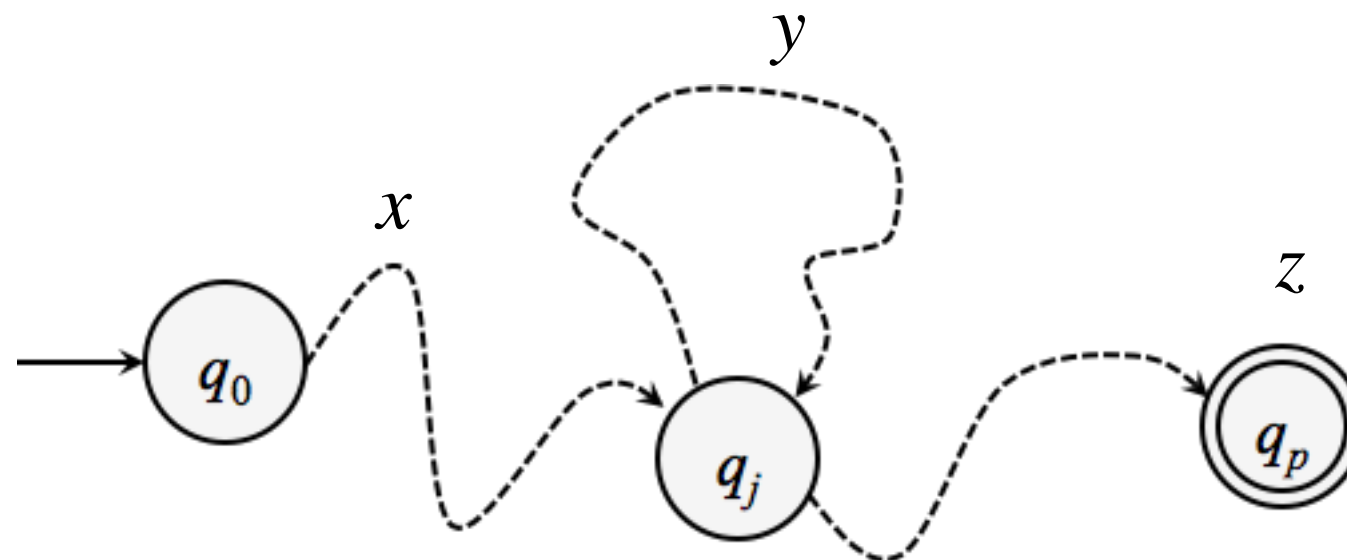
Pumping Lemma Proof

Proof. Let DFA M for L have p states. Let $w = w_1 \cdots w_n$ such that $n \geq p$ and $q_0, q_1, \dots, q_n$ be the states entered by M on w . M must revisit a state in the first p symbols. Let q_j and q_k be the first and second occurrence of this state.

Let $x = w_1 w_2 \cdots w_{j-1}$, $y = w_j w_{j+1} \cdots w_k$ and $z = w_{k+1} \cdots w_n$ which satisfies the conditions (1) and (2). Condition (3) follows from the fact that the strings xy^i are all **indistinguishable** wrt M .



PUMP ALL THE STRINGS!



Practice: Using Pumping Lemma

Problem 1. Prove that the language $L = \{a^i b^i \mid i \in \mathbb{N}\}$ is not regular using the pumping lemma.

Problem 2. Prove that $L = \{ww \mid w \in \{0,1\}^*\}$ is not regular.

Problem 3. Is the language $L = \{(ab)^i \circ (ab)^i \mid i \geq 0\}$ regular?

Problem 4. Prove that

$L = \{w \mid w \in \{0,1\}^* \text{ and } w \text{ has equal number of 1s and 0s}\}$ is not regular using pumping lemma.

Review PL Steps

- Proving L is not regular using pumping lemma
 - Assume L is regular, let p be the pumping lemma given by lemma
 - Consider a specific string $w \in L$ of length at least p such that
 - for every possible partition of w into x, y, z satisfying
 - $|xp| \leq p$ and $|y| > 0$
 - there exists an i such that $xy^iz \notin L$
- The above steps provide a contradiction to L being regular by PL
- HW 4 Problem 5
 - Show that a language is not regular and show that it satisfies conditions of the pumping lemma

Finite Automata Applications

- Lexical analysis in compilers
- Networking protocols and routing
- Circuit design and event-driven programming
- Synchronization of distributed devices

Finite Automata Limitation

- Code snippets that can only store finitely many states
- (Theory vs PL) Remember that all models of computation we discuss map directly to programming constructs

CONTAINS **11**($w[1..n]$):

```
found ← FALSE
for i ← 1 to n
    if i = 1
        last2 ←  $w[1]$ 
    else
        last2 ←  $w[i-1] \cdot w[i]$ 
    if last2 = 11
        found ← TRUE
return found
```

MULTIPLEOF5($w[1..n]$):

```
rem ← 0
for i ← 1 to n
    rem ←  $(2 \cdot \textit{rem} + w[i]) \bmod 5$ 
if rem = 0
    return TRUE
else
    return FALSE
```

